

Active Directory Domain Services (AD DS) Study Material

I. Core Concepts and Dependencies

AD DS manages users and devices, providing fundamental security services.

1. Core Services Provided by AD DS:

- **Authentication** (Kto som čo som?):
Verification of the user's identity.
- **Authorization** (Mám právo toto robiť?):
Defining what rights the user has.

2. DNS Dependency:

- **AD DS is built upon DNS.**
- AD DS cannot exist without DNS, but DNS can exist without AD DS.
- **Every domain must have at least one DNS server.**

Forest

Google.com.

Doména “.” Alebo root	root domain	tu nevidíme
Com,net,sk,ge...	top domain	
Google, telekom,...	2th layer	

Forest má domény, úplne prvá je „.” (Root)

Pod Root doménou sú Top level domény = com, edu, sk (Top domény)

Pod Top doménou sú 2nd level domény = google, tsystems (2nd level domény)

Každá táto subdoména má ďalšie subdomény a všetky sú vlastne subdomény jedného root Forestu.

Ak je jedna doména child doména druhej tak majú medzi sebou tzv. **trust**.

T.z. že z jednej môžeme vstupovať od druhej a z druhej do prvej. Tento trust je automatický a dá sa zmeniť aj na jednosmerný.

K tomuto sa hodí aj **GPO – Group Policy Object** -> GPO môžeme mať aj bez domény ale aj s V GPO môžeme nastavovať buď User veci alebo Computer veci GPO je najsilnejší nástroj na správu v rámci domén

II. Physical Components of AD DS

Physical components **handle the storage and replication of the AD database.**

Component	Description	Key Attributes
Domain Controller (DC)	Contains the read-write copy of the AD database. Holds knowledge about all objects in its domain (PCs, printers, etc.).	Recommended to have at least 2 DCs per domain for availability and redundancy.
Read-Only Domain Controller (RODC)	A special instance of a DC with a read-only database copy.	Cannot create users, change passwords, or make any changes requiring a write operation. Useful in small, potentially less secure branches (e.g., Optima) because if stolen, no changes can be made.
Global Catalogue (GC)	A special DC hosting the GC role. Contains a read copy of all objects in the forest.	Speeds up cross-forest searching. It has all objects in the forest but only their basic attributes.
Datastores	Holds AD database file, Ntds.dit, and log files.	By default, found on all DCs. in C:\Windows\NTDS + Sysvol.

Domain Controller (DC) Database:

obsahuje templaty objektov active directory.

(how the object should look like and what are his properties.)

Database = shcematics + domain objects

III. Domain

Domain tree bude ". " pod je "lab" pod je "test" to bude:

test.lab.

Test a lab budu mat DB ktore budu synchronizovane A chceme v domene mat vzdy aspon 2 DC takze ak jedna spadne nahodi sa ta druhu a DB tej 2hej DC sa sama zosynchronizuje.

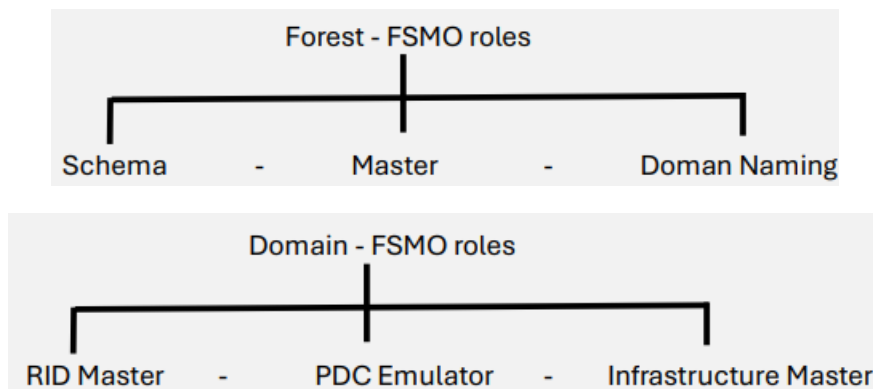
Read-only je preto ze ak ho niekto ukradne (DC) tak mu je zbytocny chceme to tak robit napr. V malych pobočkach, aby mali zamestnanci pristup

IV. Logical Components of AD DS

Logical components are **used for organization and grouping, defining security and administration boundaries**.

Component	Description	Key Relationships/Usage
Domain	First logical unit, security boundary for users, computer and other objects.	Here are the Domain Admins (DA). Often tied to the physical location of the branch.
Domain Trees	A linkage of domains that share the same root domain and DNS namespace (e.g., test. s lab.).	Domains within a tree automatically share a trust relationship, allowing entry between them. This trust can be modified to be one-way.
Forest	A collection/merging of all domains that share a common AD DS infrastructure.	Enterprise Admins (EA) administer the entire forest. The structure includes the root domain ("."), Top Level Domains (com, sk, edu), and 2nd Level Domains (google, telekom).
Site	Its a logical representation of a physical location (like a branch office or data center) that's connected by a network link.	Used to manage authentication flow; users from SITE2 should not authenticate using a DC in SITE1 due to long response times/potential login cancellation. The connection quality between sites (WAN) can be checked using PING. Sites make AD more efficient and reliable by controlling replication, authentication, and resource access based on physical network locations.
Organizational Unit (OU)	A container used for grouping objects.	Critical because it allows for delegation of permission and applying Group Policy Objects (GPOs). OUs can be structured by role (HR, dev) or location (KE, BA).
Container	Similar to an OU.	GPOs cannot be applied to Containers.

V. Flexible Single Master Operations (FSMO) Roles



FSMO (Flexible Single Master Operations) roles are **single-master roles** necessary for specific, conflict-sensitive operations in Active Directory.

Role Type	Role Name	Quantity & Location	Primary Function
Forest Roles			
	1. Schema Master	1x per Forest	Manages the template of AD objects.
	2. Domain Naming Master	1x per Forest	Manages the namespace within the forest.
Domain Roles			
	1. RID (Relative Identifier) Master	1x per Domain	Creates a unique identifier (RID) for new objects, ensuring all RIDs are unique. If a user is deleted and recreated, the new object gets a new RID and won't retain the old permissions.
	2. PDC Emulator	1x per Domain	Handles password changes and time synchronization. Time sync is crucial for Kerberos authentication.
	3. Infrastructure Master	1x per Domain	Management, and handles the connection of objects.

Kerberos Requirement:

The time difference in authentication must not exceed 5 minutes; otherwise, Kerberos will not authenticate.

VI. Group Policy Objects (GPO)

GPO is the most powerful management tool within domains.

- GPOs can configure User settings or Computer settings.
- GPOs can be used with or without a domain.

VII. Setup, Requirements, and Administration Notes

1. Hardware Requirements (VM Example):

- 2x VM Server recommended.
- vCPU: 2.
- vRAM: 4GB.
- vDisk: 50GB (A separate disk/volume is recommended for the database, Sysvol, etc.).
- vNIC: 1x.
- Licensing: 2x Standard Windows Server license.
- Applications: ADDS + DNS.

2. **Domain Naming Conventions** (To be remembered for the test):

- Maximum length: 15 characters.
- Allowed characters: Capital and small letters, numbers, underscore (_), and hyphen (-).

3. Local Administrator Account Security:

- Upon promoting a server to a DC, the local account is disabled.
- The previous local administrator account becomes the domain administrator, making it the strongest account.
- This is a security concern in large firms.

VIII. Recovery and Restoration (DSRM)

Directory Services Restore Mode (DSRM) is **used for recovery**.

- Entry:
Accessed using **F8** during boot.
- Purpose:
Allows recovery from the AD database object, specifically if SID and RID resolution fails.
- Restore Types:
 - Non-authoritative restore (default): After restoration, the DB is unsynchronized and would normally be erased.
 - Authoritative restore: Replicates specified objects from the DB onto all other DCs.

IX. Numerical and Scenario-Based Test Facts

Domain controllers fulfill 3 key roles, including:

Authentication/Authorization,

holding the database,

time synchronization.

- Redundancy Requirements (DCs):
For a Forest with 3 domains, a minimum of 6 Domain Controllers is required to ensure both availability and redundancy (3 without redundancy).
- FSMO Management (Forest Roles):
For a Forest with 10 domains, 2 Domain Controllers are needed to hold the Forest FSMO roles (Schema Master and Domain Naming Master).
- FSMO Management (All Roles Separately):
For a Forest with 10 domains, 32 Domain Controllers are needed if they must handle all FSMO roles, and no DC can hold the same role. (This is 2 Forest Roles + 10 Domains * 3 Domain Roles = 32 roles total).